CNC-Mill — Standard Operating Procedure

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> **Audience:** Maintenance Tech · Line Manager · Quality Worker
> **Applies to:** Line-A, station 2, CNC-Mill

1. Purpose

The CNC-Mill is the second station of Line-A. It receives a turned shaft blank from the CNC-Lathe and machines keyways, flats, and feature pockets into it before the part moves to the Surface-Grinder. A mill outage cascades quickly: the upstream lathe blocks within one buffer fill, and downstream grinders, deburring, and CMM all starve.

2. Normal operating envelope

- Ideal cycle time: **45 s**
- Max acceptable cycle time: **54 s**
- Expected reject rate: **≤ 3 %**
- Fault probability per cycle: **0.8 %**
- Input buffer capacity: **5 parts**

3. Idle reasons

`idle_reason`	Meaning on the CNC-Mill	First check
`Starved`	CNC-Lathe (station 1) is not feeding parts.	Inspect the lathe's status (`Fault`, `Maintenance`, or also `Idle`).
`Blocked`	Surface-Grinder (station 3) output buffer is full.	Resolve the grinder's downstream condition first.

4. Faults & corrective actions

4.1 `Spindle-Vibration`

- **Symptoms in telemetry:** `fault_type = "Spindle-Vibration"`; reject

rate trends above 3 % in the hour leading up to the fault.
- **Likely root cause:** spindle imbalance, worn tool holder, or loose drawbar.
- **Corrective action:**
  1. LOTO and isolate spindle drive.
  2. Inspect drawbar tension; re-torque to OEM spec.
  3. Replace the tool holder; rebalance the spindle assembly.
  4. Run a 30-second air-cut diagnostic and record vibration amplitude.
- **Restart criteria:** vibration amplitude < 0.5 mm/s RMS at idle.

4.2 `Coolant-Leak`

- **Symptoms in telemetry:** `fault_type = "Coolant-Leak"`; operators may also report a visual puddle around the enclosure.
- **Likely root cause:** failed enclosure seal, cracked hose, or loose flood-coolant fitting.
- **Corrective action:**
  1. LOTO and drain the coolant tank.
  2. Trace the leak; replace the failed seal, hose, or fitting.
  3. Refill coolant to spec concentration (5–8 % emulsion).
  4. Run a 5-minute leak test before resuming.
- **Restart criteria:** no visible leak across a 5-minute idle test.

4.3 `Tool-Wear`

- **Symptoms in telemetry:** `fault_type = "Tool-Wear"`. May be preceded by rising actual cycle time and increasing reject rate, particularly with surface finish callouts at CMM-Inspection.
- **Likely root cause:** insert end of service life; aggressive feed.
- **Corrective action:**
  1. Index the worn tool out of service.
  2. Install fresh insert; reset tool offset against the probe.
  3. Re-inspect any parts machined in the prior 10 cycles at CMM.
- **Restart criteria:** tool offset within 0.005 mm of master.

5. Preventive maintenance schedule

- **Daily:** check coolant concentration; clear chip evacuation auger.
- **Weekly:** spindle drawbar torque check; air-blow tool changer.
- **Monthly:** way-cover inspection; replace coolant filter.
- **Quarterly:** spindle vibration analysis; ATC mechanical service.

6. References

- `Maintenance_Workflow_Overview.pdf`
- `Safety_Lockout_Tagout_Procedure.pdf`
- `Quality_Reject_Disposition_Policy.pdf`
- `Line-A_01_CNC-Lathe_SOP.pdf`
- `Line-A_03_Surface-Grinder_SOP.pdf`